

# Nathaniel Billig

815-245-2448 | nbillig@uwyo.edu | <https://nbillig.github.io/> | [linkedin.com/in/nathaniel-billig/](https://www.linkedin.com/in/nathaniel-billig/)

## EDUCATION

**University of Wyoming**, College of Engineering and Physical Sciences

**Expected May 2027**

Master of Science: Electrical Engineering

Cumulative GPA 3.9/4.0

**Relevant Coursework:** *Advanced Robotics Modeling, Robot Learning, Robot Perception, Cooperative Robotics, AI for Multiagent Systems*

**University of Wyoming**, College of Engineering and Physical Sciences

**May 2025**

Bachelor of Science in Computer Engineering; Minors in Computer Science and Mathematics Cumulative GPA 3.8/4.0

**Relevant Coursework:** *Advanced Microprocessors, HDL Digital Design, Electronics, GPU Programming for HPC, Computer Security, Neural Networks, Deep Reinforcement Learning*

## EXPERIENCE

**Robotics Researcher — USSF SSTI Sponsored Research**

**Sept 2024 - Present**

**University of Wyoming**

- Designed and built a three-finger, tendon-driven robotic gripper for USSF-sponsored research, including CAD modeling, sensor selection and integration, and embedded software.
- Designed a four-layer PCBA in Altium CircuitMaker for an ESP32-based controller, incorporating motor drivers, Hall-effect sensor interfaces, encoders, and load-cell amplifiers, taking the board from schematic design through layout and submission for manufacturing.
- Tested gripper prototypes and revised finger geometry, sensor placement, and mechanical design based on grasping and sensing limitations and sponsor feedback.
- Developed simulation and calibration methods to characterize gripper kinematics, joint sensing, and contact forces

**Microcontroller Project Engineer, 4th State Communications - Cheyenne, WY**

**Oct 2023 - Feb 2024**

- Developed C/C++ Arduino firmware for UART communication and control of telecommunications transmitters
- Built Python software using Linux NFS to transmit control commands between a central controller and networked relay systems

**Undergraduate Research Assistant, University of Wyoming - Laramie, WY**

**May 2023 - Sept 2025**

- Developed computer vision applications and embedded Python/C/C++ software for Arduino and Raspberry Pi-based autonomous platforms
- Instructed high school students and teachers in physical computing, programming fundamentals, and hands-on electronics

## PROJECTS

**Hall Effect Robotic Joint Angle Sensing**, University of Wyoming

· Embedded magnets and Hall-effect sensors in gripper joints to measure joint angles. Tested sensor placement and developed calibration curves and embedded software for real-time measurements.

**Multi-Agent Continuum Arm Q-Learning**, University of Wyoming

· Developed a multi-agent Q-learning framework for coordinated control of a continuum robotic arm, evaluating discrete state and action spaces for joint coordination and motion control

**Robotic Gripper Modeling**, University of Wyoming

· Built a MATLAB/Simscape model of the gripper, compared simulated joint motion with experimental measurements, and estimated contact forces during grasping.

## HONORS & AWARDS

**Crowd Favorite Poster — USSF University Consortium Symposium, 2026**

**Computer Engineering Student of the Year — Wyoming Engineering Society, 2025**

**Tau Beta Pi Engineering Honor Society — Inducted 2024**

## SKILLS

**Programming & Embedded Systems:** C/C++, Embedded Programming, Python, MATLAB, Arduino, Raspberry Pi, ESP32

**Electronics & Communication:** UART, SPI, I<sup>2</sup>C, PCB/Schematic Design, Sensor Integration, Soldering

**Robotics & Simulation:** ROS, Kinematics, Path Planning, Digital Twins, Reinforcement Learning, Computer Vision

**Software & CAD:** SolidWorks, Fusion 360, CircuitMaker, OpenCV, MATLAB/Simulink, Simscape, Microsoft Office Suite